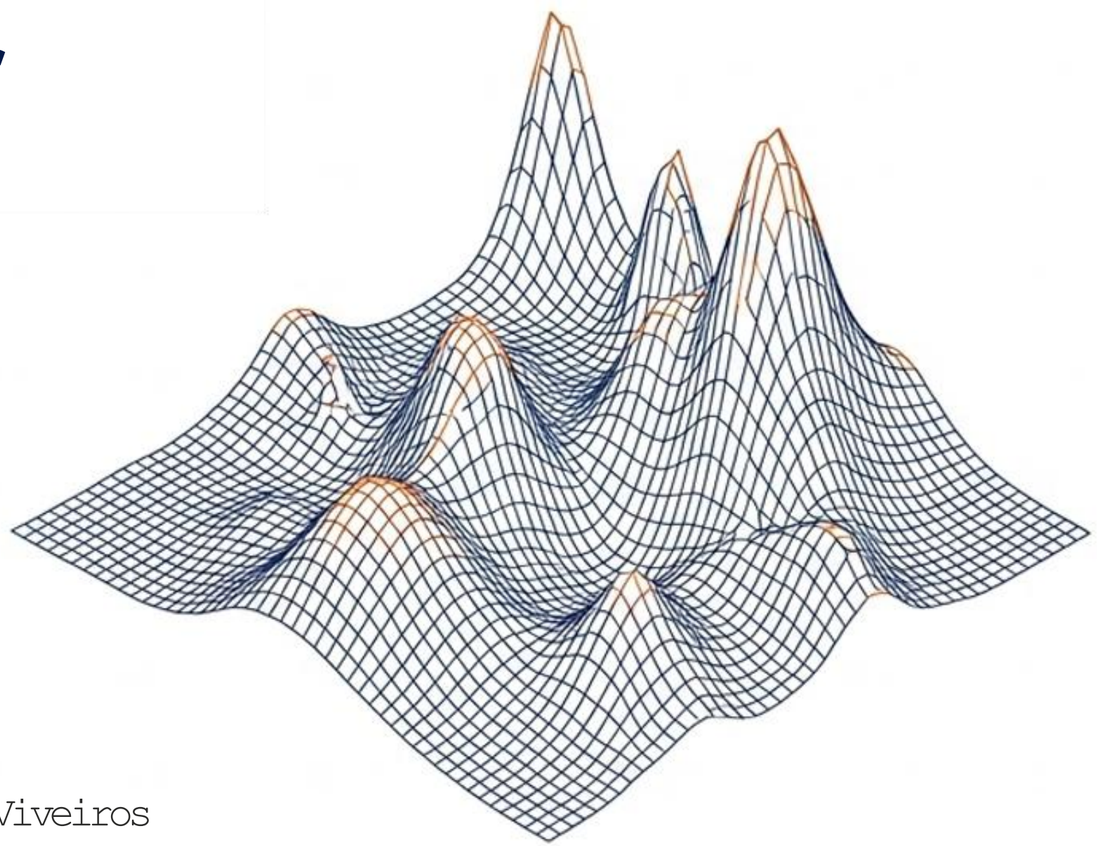
Forecasting Next-Week Realized Volatility for S&P 500 Sector and Industry ETFs

A TensorFlow-Based Approach Using FactSet Data and Subgroup Evaluation



COURSE: DATASCI 207 Final Project Proposal
TEAM: Vinit Bhatt, Siddharthen Sridhar, Ronald Ho, Diogo Viveiros
STATUS: Project Proposal

Motivation: Volatility is the Driver of Portfolio Risk and Drawdowns

The Problem

Short-horizon returns are often random noise, but volatility exhibits persistence and clustering. When volatility spikes, correlations converge to 1.0, and risk controls fail.

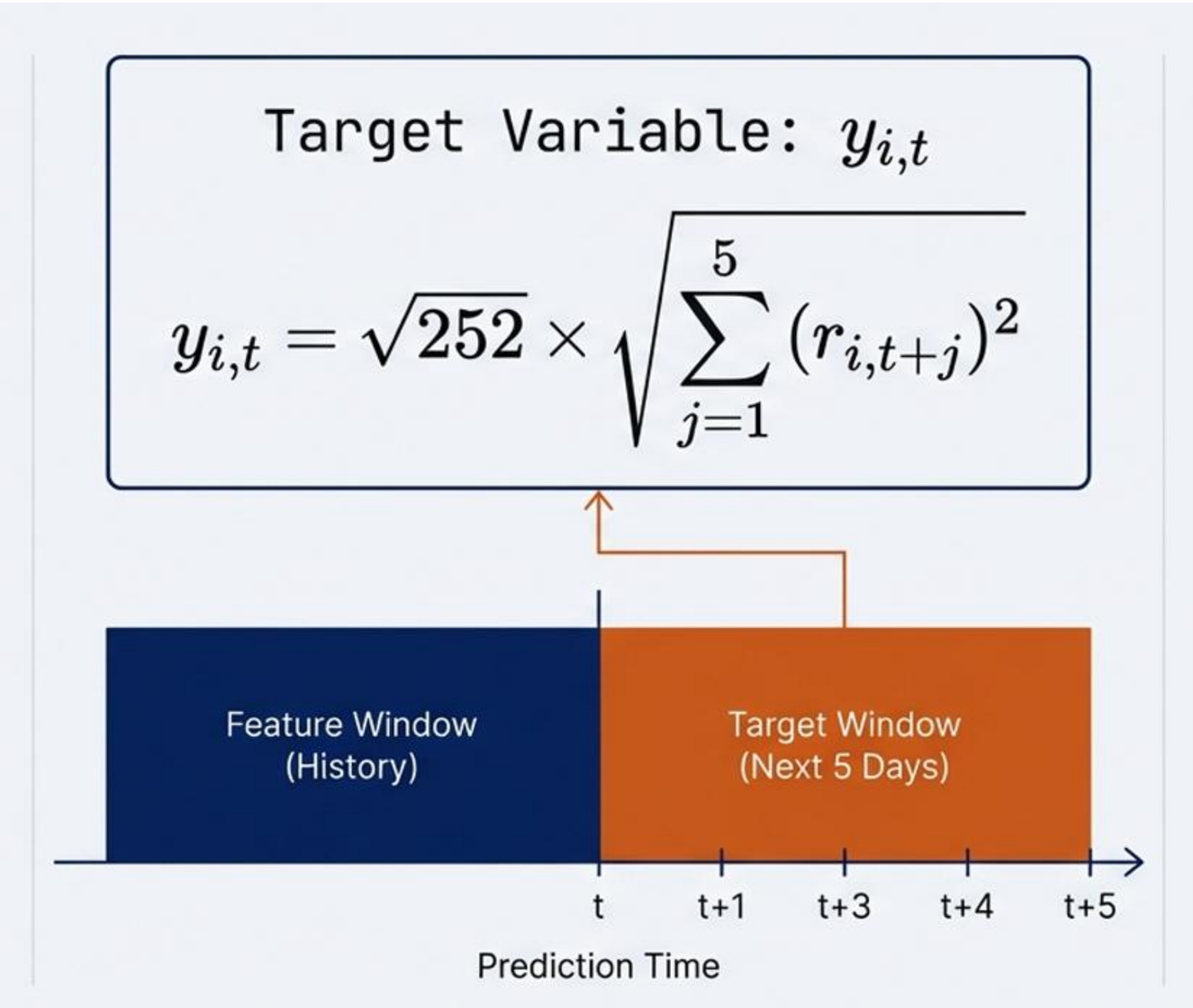
The Opportunity

Unlike direction (up/down), volatility is a predictable target. Forecasting next-week volatility allows for pre-emptive position sizing and risk budgeting before the damage is done.

Problem Definition: The Math Behind the Prediction

The Target

- Annualized Realized Volatility.
- Horizon: Next 5 Trading Days.
- Scope: Continuous regression problem (not binary classification).



Key Constraints

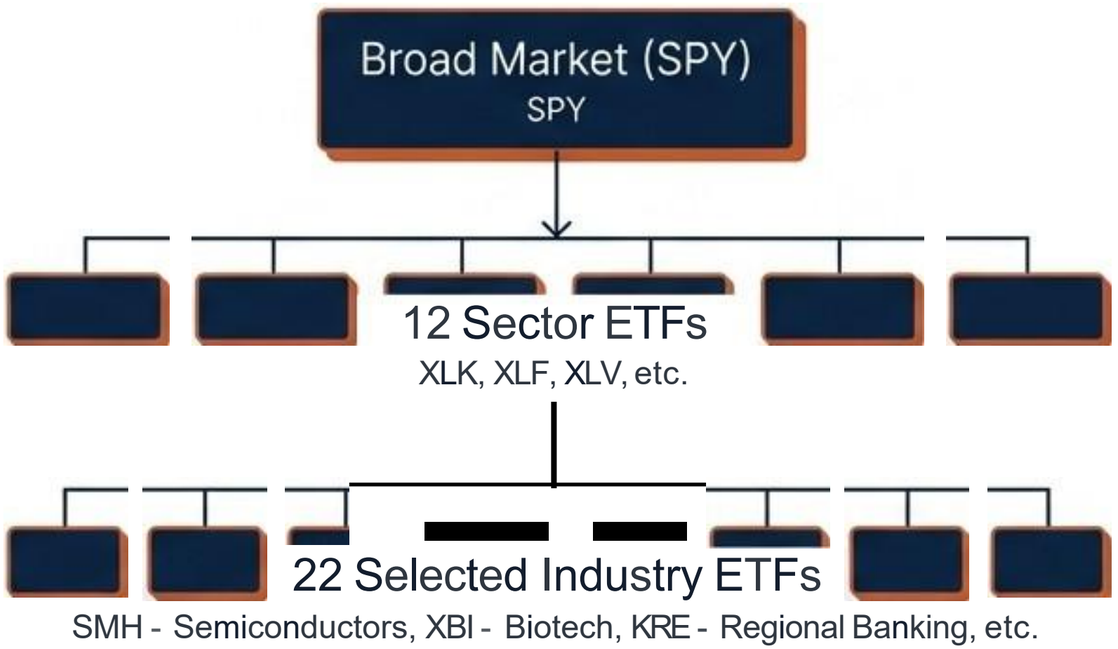
- No Look-Ahead Bias: Predictions at time t use only features available at or before t.
- r represents Log Returns.

The Data Advantage: Professional-Grade Universe via FactSet

Data Source & Quality

- Utilizing university access to FactSet, a professional investment platform ($20k/yr value). This ensures adjusted pricing accuracy compared to free data sources.
- Timeline: Daily data from 2015-2025.
- Core Fields: Adjusted Closing Price, Trading Volume.

The Universe



Universe Granularity allows for specific Subgroup Analysis (Sector vs. Industry).

Feature Engineering: Extracting Signal from from Price and Volume

Rationale: Volume shocks often reflect information arrival that precedes price volatility.

Modeling Strategy: From Heuristics to Neural Networks

The Baseline

Volatility Persistence Heuristic
Inter

Rule-based model assuming next-week volatility equals trailing 20-day volatility.

Role
The Benchmark. ML models must beat this to be considered useful.

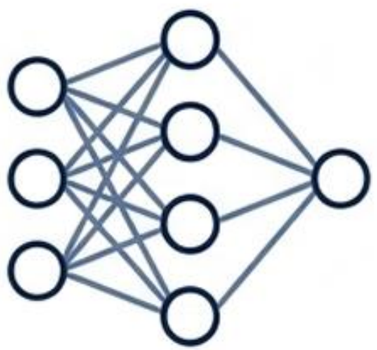
Linear Model

TensorFlow Linear Regression

Establishes baseline feature importance and linear relationships between volume/trend and future volatility.

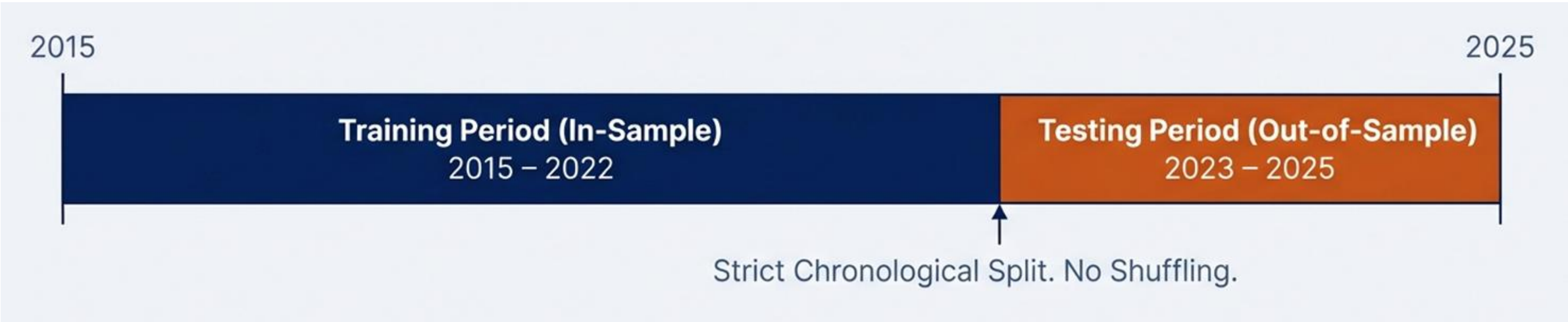
Deep Learning

TensorFlow Feed-Forward NN (MLP)



Multi-Layer Perceptron designed to capture non-linear interactions. End-to-end pipeline built in TensorFlow.

Evaluation Protocol: Realistic Time-Series Splitting



Primary Metrics:

1. **MAE** (Mean Absolute Error): Measures average **magnitude** of forecast error.
2. **RMSE** (Root Mean Square Error): Penalizes **large outliers**, critical for risk management contexts.

Error Analysis:

Visual inspection of Prediction vs. Actuals to identify **failure modes** during **market shocks**.

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Subgroup Analysis: Sector vs. Industry Performance

The Hypothesis

Are broad Sectors (e.g., Technology) easier to forecast than specific Industries (e.g., Semiconductors)? This

analysis informs portfolio construction-placing higher confidence in models for stable subgroups.

Reporting Levels

Level 1: Global Metrics

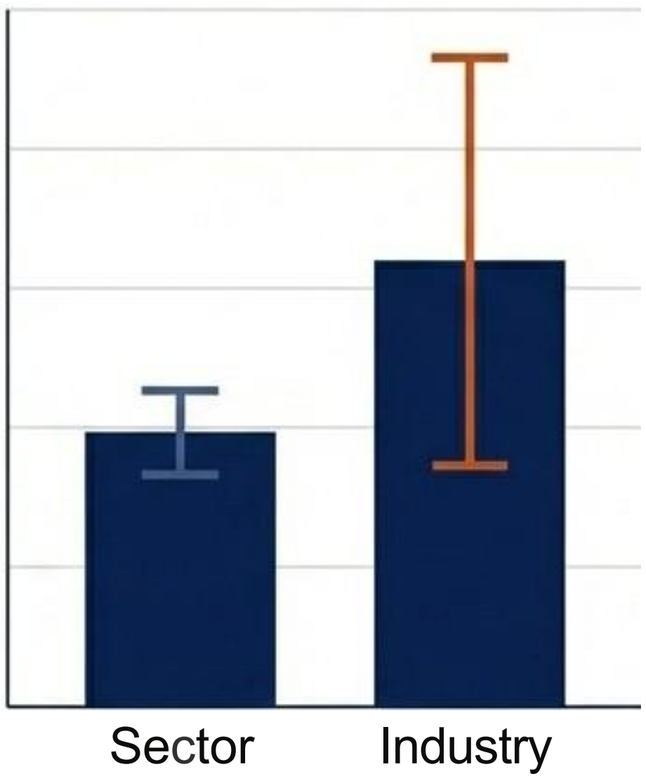
- Overall MAE/RMSE across the entire SPY universe.

Level 2: Subgroup Metrics

- Aggregated performance for 12 Sector ETFs vs. 22 Industry ETFs.

Level 3: Idiosyncratic Analysis

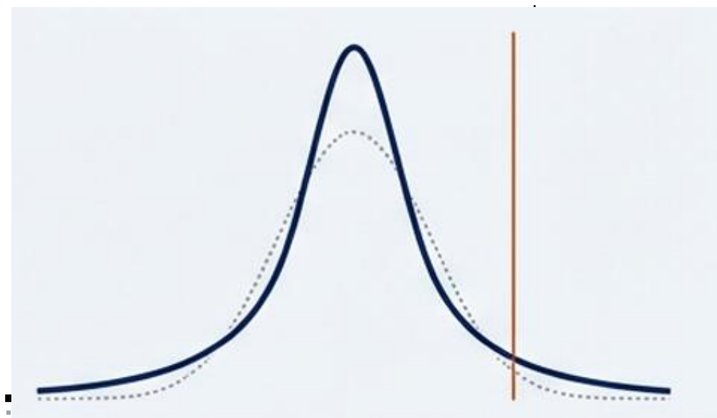
- Identification of specific industries that are systematically harder or easier to forecast.



Forecast Error Variance

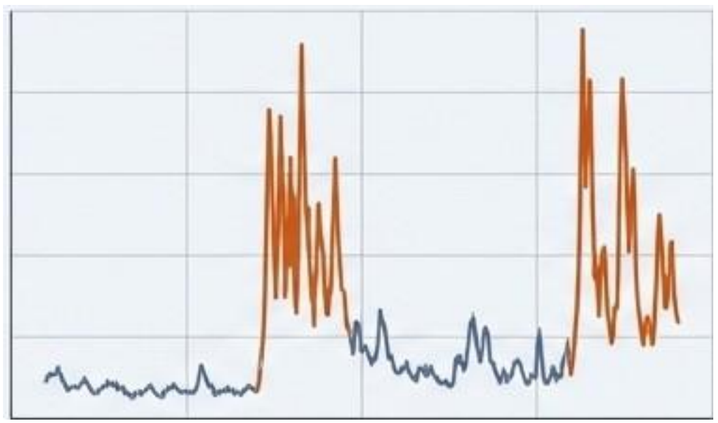
Planned Exploratory Data Analysis (EDA)

Distributions & Fat Tails



Analyzing price/return distributions for non-normality.

Volatility Clustering



Visualizing persistence: high volatility is followed by high volatility.

Correlations & Regimes

Heatmaps showing how Sector correlations tighten during stress periods.

Data Integrity Checks

- Missing data verification in FactSet pull.
- Ticker mapping validation across 2015-2025.

Limitations and Risks

Overfitting

Neural networks may memorize noise in the training set (2015-2022) rather than learning generalizable patterns.

Non-Stationarity

Market structural changes (regime shifts) in the 2023-2025 test set may differ fundamentally from the training period dynamics.

Black Swan Events

Models trained on 'normal' volatility regimes may fail to predict extreme outliers or unprecedented market shocks.

Universe Bias

The selected ETF universe is a proxy for the economy but may not perfectly represent all market conditions.

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Project Deliverables and Roadmap

The Deliverables

- **GitHub Repository:** Full source code with reproducible notebooks.
  - **Notebook 1:** Data Prep & EDA
  - **Notebook 2:** Feature Engineering
  - **Notebook 3:** Modeling & Evaluation
- **Reports:** Milestone Report & Final Report
- **Presentation:** In-class findings presentation

Execution Plan

Team: Vinit S. Bhatt, Siddharthen Sridhar, Ronald Ho
Status: Data extraction (FactSet) and Proposal Drafting complete.

Data Extraction (Done) | **EDA& Features** | **Model Training** | **Final Analysis**

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Summary: From Prediction to Risk Management

THE PROBLEM

Volatility destroys value. We need to see it coming.

THE SOLUTION

A TensorFlow Neural Network pipeline trained on professional FactSet data.

THE VISION

To build a 'risk radar' superior to historical averages, enabling safer portfolio construction.

Thank You. Questions?

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